

JINKYU LEE, PH.D.

AI / LLM / NLP Researcher · LLM Solutions Architect · Founder, HappyAI

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PROFESSIONAL SUMMARY

Ph.D.-level AI/NLP researcher, LLM solutions architect, entrepreneur, and technical educator with extensive experience in domain-specific LLM development, RAG systems, embedding-based retrieval, parameter-efficient fine-tuning, document intelligence, and responsible AI evaluation. Founder of HappyAI, leading client-facing AI projects for government agencies, public institutions, enterprises, universities, and research organizations in Korea.

Doctoral research developed a structure-aware bias evaluation and mitigation framework for transformer-based language models, comparing encoder-based BERT classification models and decoder-based GPT-series generative models. Experienced in translating experimental NLP and LLM workflows into production AI systems — private LLM platforms, RAG-based document intelligence, structured information extraction, semantic search, and LLM-powered text analytics SaaS.

CORE STRENGTHS FOR NVIDIA AI TECHNOLOGY CENTER

Foundation Model / LLM Application	Private LLM, RAG, document intelligence, structured extraction, domain-specific LLM workflows, and commercial LLM SaaS products.
Fine-Tuning & Adaptation	LoRA / QLoRA, PEFT, instruction-tuning, BERT / KoPatBERT fine-tuning, and domain-adaptation workflows.
Evaluation & Responsible AI	Ph.D. research on BERT/GPT bias measurement, IHTA mitigation, CoT/IAT-inspired prompting, Likert/ordinal/interval scoring, and multi-agent self-refinement.
Customer & Research Collaboration	Founder and technical lead delivering 50+ AI projects across public sector, industry, universities, and research institutes.
Technical Communication	Adjunct professor, enterprise AI instructor, columnist, and book author translating frontier AI concepts for researchers, executives, and practitioners.

TECHNICAL SKILLS

LLM / Foundation Model Applications: Domain-specific LLM applications, private LLM, RAG, document intelligence, instruction-following extraction, structured generation, prompt orchestration, evaluation-driven improvement.

Fine-Tuning / Adaptation: Supervised fine-tuning, LoRA, QLoRA, PEFT, Hugging Face Transformers / Datasets / Accelerate, Sentence Transformers, BERT / KoBERT / KoPatBERT, GPT-style decoder models.

RAG / Retrieval / Embedding: Document ingestion, chunking, embedding generation, vector search, reranking, semantic search, retrieval evaluation, hallucination reduction, domain-specific QA.

Evaluation / Responsible AI: Bias measurement, fairness evaluation, demographic bias analysis, CoT prompting, IAT-inspired prompts, Likert/ordinal/interval scoring, answer relevance, factuality, format consistency.

NVIDIA-Relevant Ecosystem: NVIDIA AI software stack concepts — Megatron-LM, Megatron-Bridge, NeMo / NeMo-RL-style post-training, TensorRT-LLM-style inference optimization, CUDA / GPU-aware training and serving.

Programming / Analytics: Python, PyTorch, LangChain, NLP pipelines, text mining, topic modeling, sentiment analysis, clustering, classification, crawling, survey and media analytics.

PROFESSIONAL EXPERIENCE

Founder / LLM Solutions Architect — HappyAI

Jul 2023 – Present

Seoul, South Korea

- Founded and lead an AI and big-data analytics company specializing in LLM, RAG, NLP, text mining, and document intelligence solutions for public institutions, enterprises, universities, and research organizations.
- Design and deliver customized AI solutions including private LLM systems, RAG-based knowledge search, LLM-powered text analytics, structured document extraction, and domain-specific AI workflows.
- Lead end-to-end client-facing AI projects from requirements analysis and architecture design through implementation, evaluation, delivery, training, and stakeholder communication.
- Built and commercialized **DocuNova AI SaaS** (private document analysis, summarization, RAG search, report management) and designed **TexNova**, an LLM-powered text big-data analytics platform.
- Delivered LLM, RAG, fine-tuning, and AI application development training for enterprise and public-sector audiences including Samsung SDS, KT, and Seoul Digital Foundation.

Adjunct Professor, Dept. of AI Software — Hansung University

Feb 2026 – Present

Seoul, South Korea

- Teach practical AI software, NLP, LLM, RAG, prompt engineering, fine-tuning, and AI application development.
- Develop practice-oriented curricula connecting LLM theory, transformer architectures, RAG, LoRA/QLoRA fine-tuning, and real-world AI service implementation.

Generative AI Columnist — Public News

Dec 2023 – Dec 2025

Korea

- Published public-facing columns on LLM limitations, RAG chatbots, AI bias, gender and religious bias, explainability, and Chain-of-Thought prompting.
- Translated technical AI risks and mitigation concepts into accessible language for a broad public audience.

AI Researcher — Stellavision

Jul 2021 – Jul 2023

Korea

- Developed AI and big-data analytics solutions using NLP, text mining, crawling, machine learning, BERT-based modeling, Word2Vec, topic modeling, sentiment analysis, and network analysis.
- Built analytical pipelines for survey data, news, app reviews, academic papers, social media, and public-sector text across healthcare, education, legal, policy, environment, and consumer domains.

Big Data Analyst — Textom

Oct 2021 – Dec 2022

Korea

- Conducted text mining and big-data analytics projects on media, social, academic, and public datasets; delivered reports on public perception, policy issues, consumer trends, and cultural content.

Researcher, Big Data Analytics & NLP — National Institute of Forest Science

Sep 2018 – Mar 2021

Korea

- Applied NLP, text mining, GIS data analysis, survey analytics, and media analytics to forest recreation, public demand forecasting, and policy planning research.
- Contributed to research reports, policy support materials, academic publications, and press releases with government researchers and academic experts.

SELECTED NVIDIA-RELEVANT PROJECTS

Private LLM & RAG System for Enterprise Technical Documents

2024 – Present

Korea Electric Power Engineering & Construction

- Designed a private LLM-based document intelligence system for secure enterprise knowledge search and QA.
- Built a RAG pipeline: document ingestion, chunking, embedding generation, vector retrieval, prompt orchestration, and grounded answer generation.
- Applied domain-specific retrieval to improve factual grounding and reduce hallucination in technical QA; explored LoRA/QLoRA domain adaptation and quantization-aware deployment for efficient private LLM operation.
- Evaluated answer relevance, retrieval quality, and response consistency using domain-specific test cases.

LLM-Based Legal Judgment Information Extraction System

2023

Ministry of Justice

- Developed an LLM-based legal document understanding system to extract structured crime-fact elements from Korean court judgments.
- Designed instruction-following prompts, structured output schemas, and validation workflows; built preprocessing for long documents (segmentation, context management, schema-based generation).
- Investigated instruction tuning and domain adaptation to improve output consistency and legal-terminology handling.

Instruction Fine-Tuning & RL-Based LLM Adaptation Research

2023

Seoul Digital Foundation

- Conducted applied research on LLM adaptation using instruction fine-tuning and RL-based post-training concepts.
- Designed instruction-response datasets and evaluation workflows for task-specific behavior alignment; studied RLHF/RLAIF-style pipelines aligned with modern frameworks such as NVIDIA NeMo-RL.

DocuNova AI SaaS — Private Document Intelligence Platform

2025 – Present

HappyAI

- Designed and developed a private document intelligence SaaS for automatic analysis, summarization, RAG search, and large-scale report management.
- Built end-to-end RAG workflows (parsing, chunking, embedding, vector search, prompt construction, grounded generation) with retrieval evaluation and prompt refinement to improve faithfulness and domain relevance.

LLM & Embedding-Based Software Company Taxonomy

2025

Software Policy & Research Institute

- Developed an LLM and embedding-based framework for structuring software company information using semantic embedding, clustering, and classification.
- Used LLM-assisted interpretation to convert large-scale text-mining results into explainable taxonomy insights.

KoPatBERT-Based Patent & Academic Paper QA / Classification

2023

Korea Technology Market

- Fine-tuned KoPatBERT transformer models for patent and academic-paper classification and QA; built Korean technical-document pipelines and evaluated performance via task-specific validation and error analysis.

AI-Based Road Maintenance Document Automation

2025

National Construction / Infrastructure Research Institution

- Built an AI document automation system to standardize road-maintenance input and convert unstructured descriptions into structured forms, with schema-constrained generation and validation logic.

DOCTORAL RESEARCH

Dongguk University — Ph.D. in Artificial Intelligence · A Study of BERT Classification Models and GPT-Series Generative Models

- Built a structure-aware LLM evaluation framework to analyze bias across transformer encoder and decoder architectures, using BERT classification models and GPT-series generative models as representative systems.
- For encoder models: a BERT classification evaluation pipeline with demographic-term masking, hierarchical generalization, threshold-based augmentation, and IHTA-based mitigation.
- For decoder models: prompt-based evaluation protocols using Chain-of-Thought prompting, IAT-inspired association tests, decision-bias prompts, and Likert/ordinal/interval-scale scoring to quantify implicit bias.
- Proposed lightweight, retraining-free mitigation (neutral instruction insertion, multi-agent self-refinement) suitable for production LLM evaluation and governance — connecting AI fairness research to enterprise and public-sector AI deployment.

EDUCATION

Ph.D. in Engineering, Artificial Intelligence — Dongguk University

Feb 2022 – Feb 2026

Focus: transformer-based language models, LLM bias, fairness, evaluation, BERT/GPT architectures, NLP, responsible AI.

M.A. in Recreation, Welfare and Culture — Kookmin University

Sep 2016 – Aug 2018

B.Eng. — Kookmin University

Feb 2012 – Aug 2016

SELECTED PUBLICATIONS

- Measuring and Mitigating Implicit Bias in Large Language Models. *Connection Science*, 2026.
- Improving Commonsense Bias Classification by Mitigating the Influence of Demographic Terms. *IEEE Access*, 2024.
- Big Data-Based Analysis of Domestic AI Fairness Research Trends Using KERIS Papers. *Journal of Korea Multimedia Society*, 2025.
- Analysis of Technical Perception of Large Language Models Using News Article Big Data: Before and After ChatGPT. *Journal of Korea Multimedia Society*, 2024.
- Improving Generation of Sentiment Commonsense by Bias Mitigation. *Journal of Korea Information Science Society*, 2023.
- Plus multiple KCI-indexed publications on NLP-based text mining, research-trend analysis, healthcare/social-media analytics, forest-recreation big data, and media-perception analysis.

BOOKS, TEACHING & TECHNICAL COMMUNICATION

- **Author** — *Domain-Specific Custom LLM Guide* (manuscript completed): practical guide covering LLM concepts, transformer architectures, RAG, fine-tuning, LoRA/QLoRA/PEFT, Hugging Face workflows, GPU-based training, and domain-specific LLM system design.
- **Author** — *Stock Data Analysis with ChatGPT* (Youngjin.com): introduction to Python and ChatGPT-assisted stock data analysis.
- **Enterprise / Public-Sector AI Instructor** — delivered training on LLM fundamentals, RAG, LangChain chatbots, LoRA/QLoRA fine-tuning, LLM/sLLM application development, NLP, prompt engineering, and Python data analysis for Samsung SDS, KT, Inflearn, Seoul Digital Foundation, and others.
- **Industry-Academia Collaboration** — Dongguk University Knowledge Science Research Center: internship program partner through HappyAI supporting AI/big-data technology exchange, joint project planning, and research collaboration.